Pseudodifferential Operators

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1 Differential Operators

The point of these notes is to give a brief introduction to pseudodifferential operators. First we will define ordinary differential operators on manifolds, which will give context and motivate what is to come. Most of these notes are from [1]

First, we will look at a very basic example of a differential operator: The Cauchy-Riemann operator on $\mathbb{C}$. Let $f : \mathbb{C} \to \mathbb{C}$ be a smooth function. Writing $f(z) = u(z) + iv(z)$, we say f is holomorphic if

$$\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = 0 \tag{1}$$

$$\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = 0, \tag{2}$$

with the obvious identification of $z = x + iy$. Continuing to think of f as a map $\mathbb{R}^2 \to \mathbb{R}^2$, we want to think of the left side of the equation as a differential operator

D applied to $f = (u(x,y), v(x,y))$. Notice in order to map $\mathbb{R}^2$ valued functions to $\mathbb{R}^2$ valued functions, D must have matrix valued coefficients. We can write

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \frac{\partial}{\partial x} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \frac{\partial}{\partial y},$$

so that indeed

$$Df = \begin{bmatrix} \frac{\partial u}{\partial x} & 0 \\ 0 & \frac{\partial v}{\partial x} \end{bmatrix} + \begin{bmatrix} 0 & -\frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial x} & 0 \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} & -\frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial x} \end{bmatrix}.$$

In other words, D is a linear map $C^\infty(\mathbb{R}^2, \mathbb{R}^2) \to C^\infty(\mathbb{R}^2, \mathbb{R}^2)$. This suggests that instead of considering differential operators acting on functions (sections of a trivial line bundle,) we consider differential operators which act on sections of vector bundles.

In the following section, we let X be a manifold, with E and F smooth complex vector bundles on X. We denote by $\Gamma(E)$ the space of smooth C^∞ sections of E.

Definition 1.1. A *differential operator* of order m on X is a linear map $P : \Gamma(E) \to \Gamma(F)$, such that if we choose local coordinates x_i in an open set $U \subset X$, and also local trivializations $E|_U \simeq U \times \mathbb{C}^p$ and $F|_U \simeq U \times \mathbb{C}^q$, then P has the form

$$P = \sum_{|\alpha| \leq m} A^\alpha(x) \frac{\partial^{|\alpha|}}{\partial x^\alpha}$$

where the A^α's are smooth $p \times q$ matrices of smooth functions, and such that one of the highest order coefficients is nonzero somewhere.

Observe that if we make a change of local trivializations of $E|_U$ and $F|_U$ by maps $g_E : U \to GL_p(\mathbb{C})$ and $g_F : U \to GL_p(\mathbb{C})$, then in these trivializations, P has the form

$$P = g_F \left(\sum_{|\alpha| \leq m} A^\alpha \frac{\partial^{|\alpha}}{\partial x^\alpha} \right) g_E^{-1} = \sum_{|\alpha| \leq m} \tilde{A}^\alpha \frac{\partial^{|\alpha|}}{\partial x^\alpha},$$

where the coefficients $\tilde{A}^\alpha$ can be obtained using the product rule to interchange the orders of ∂_x and g_E^{-1}. Namely, for all $|\alpha| = m$, we have the formula

$$\tilde{A}^\alpha = g_E A^\alpha g_E^{-1}$$

for the modified coefficients.

Similarly, if we change to coordinates $\tilde{x} = \tilde{x}(x)$ on the set U, P takes the form

$$P = \sum_{|\alpha| \leq m} \tilde{A}^\alpha(\tilde{x}) \frac{\partial^{|\alpha|}}{\partial x^\alpha}.$$

Using the Jacobian for the change of variables, for any j we can write

$$\frac{\partial}{\partial x_j} = \sum_{k=1}^{n} \frac{\partial \tilde{x}_k}{\partial x_j} \frac{\partial}{\partial \tilde{x}_k}.$$

For any multi-index α with $|\alpha| = m$, we can write

$$\frac{\partial}{\partial x^\alpha} = \sum_{|\beta|=m} \left[\frac{\partial \tilde{x}}{\partial x}\right]_\beta^\alpha \frac{\partial}{\partial \tilde{x}^\beta},$$

where $\left[\frac{\partial \tilde{x}}{\partial x}\right]_*^*$ denotes the symmetrization of the m-th exterior power of the jacobian. Note that both upper and lower indices are symmetrized to reflect the commutativity of partial derivatives.

Thus we have

$$\tilde{A}^\alpha = \sum_{|\beta|=m} A^\beta \left[\frac{\partial \tilde{x}}{\partial x}\right]_\beta^\alpha$$

for the top order coefficients with $|\alpha| = m$.

We see that P behaves well under coordinate changes and changes of trivialization, so long as we only consider the top order terms. In fact, we have proven that the coefficients $\{i^m A^\alpha\}_{|\alpha|=m}$ represent a well defined section $\sigma(P)$, called the *Principal Symbol*, of the bundle $(\odot^m TX) \otimes \mathrm{Hom}(E, F)$, where $\odot$ denotes the symmetric tensor product.

Next we note that the space $\odot^m V$ is isomorphic to the space of homogeneous polynomial functions of degree m on V^*. Hence, for each cotangent vector $\xi \in T_x^* X$, the principal symbol gives a linear map

$$\sigma_\xi(P) : E_x \to F_x.$$

In other words, the symbol can be considered as a (polynomial) function on the cotangent bundle.

1.0.1 Examples on Riemannian Manifolds

Now let X be a riemannian manifold equipped with a metric $\langle \cdot, \cdot \rangle$. In local coordinates we can express the metric using the basis as

$$\sum g_{jk} dx_j dx_k$$

for some matrix g_{jk}. We also denote by g^{jk} the inverse of g_{jk}, and $g := \det(\{g_{jk}\})$.

Example. Consider the Laplace-Beltrami operator $\Delta : C^\infty(X) \to C^\infty(X)$ defined by $\Delta = d^* d$, where d^* is defined on k-forms α by $d^*\alpha = (-1)^{(nk+n+1)} * d^*\alpha$. We first calculate an explicit formula for d^* applied to k-forms $\alpha = \sum A_i dx^i$:

$$d^*\alpha = \frac{1}{\sqrt{g}} \sum_{ij} \frac{\partial}{\partial x_j}(A_i g^{ij} \sqrt{g}).$$

Then we can apply it to $\alpha = df = \sum \frac{\partial f}{\partial x_i} dx^i$ to obtain the following formula for Δ:

$$\Delta f = \frac{1}{\sqrt{g}} \sum_{ij} \frac{\partial}{\partial x_j}(\sqrt{g} g^{ij} \frac{\partial f}{\partial x_i}) \tag{3}$$

$$= \sum_{j,k=1}^{n} g^{jk} \frac{\partial^2 f}{\partial x_j \partial x_k} + \{\text{lower order terms}\}. \tag{4}$$

Now we can calculate its symbol: for some cotangent vector $\xi = \sum \xi_k dx^k$, the polynomial function is given by

$$\sigma_\xi(\Delta) = i^2 \sum_{j,k=1}^{n} g^{jk} \xi_j \xi_k = -\|\xi\|^2.$$

Example. From this discussion, it is natural to wonder if we can just define an operator with symbol $i\|\xi\|$. In fact we could, but it will be pseudodifferential and not quite what we are after in several ways.

Instead, we introduce the concept of clifford action of T^*M on a bundle S. In defining the dirac operator D in the usual way, we see that its symbol is

$$\sigma_\xi(D) = ic(\xi),$$

where $c(\xi) \in \text{End}, mk(S)$ denotes clifford multiplication by ξ. Trivializing S in some neighborhood U, $c(\xi)$ becomes some matrix-valued function on U, and thus D is an ordinary differential operator on S.

Remark. For a differential operator $P : \Sigma(E) \to \Sigma(F)$ we can consider $\sigma(P)$ as a bundle map of the pullbacks $\pi^*E \to \pi^*F$ in the natural way, under the projection $\pi : T^*M \to M$.

2 Sobolev Spaces (L^2 Based)

In the last section, we defined differential operators on manifolds, mapping between sections of vector bundles. In the later section, we will define a more general class of operators called Pseudodifferential operators. These are a natural generalization of ordinary differential operators, as we will see, with their own corresponding principal symbols. These operators arise naturally when we want to find approximate inverses of differential operators, called *parametrices*, which are inverses modulo a very nice class of operators called smoothing operators.

But first, we need to define the spaces of functions these operators act on. We require them to be continuous and, say, C^∞, at least at first. We will try to base a norm on the L^2 norm because of its nice duality properties and so that we can eventually consider non-smooth functions in our domains. The ordinary L^2 norm is not good enough, however. Consider the sequence of functions

$$u_n(x) = x^n$$

on the interval $[-1, 1]$. Since

$$\|u_n\|_{L^2} \to 0$$

as $n \to \infty$, but

$$\|u_n'\|_{L^2} = n,$$

there may not exist a $C > 0$ such that $\|u'\|_{L^2} \leq C\|u\|_{L^2}$.

However, define a norm $\|\cdot\|_1$ on $\mathbb{R}$ given by

$$\|u\|_1 = \|u\|_{L^2} + \|u'\|_{L^2}.$$

If we consider $D = \frac{\partial}{\partial x}$ as a map from $C_c^\infty(\mathbb{R})$ with the L^2 norm to the space $C_c^\infty(\mathbb{R})$ with the $\|\cdot\|_1$ norm, D becomes a bounded operator. By taking one derivative, we lose one order of control over the "size" of our function than before. In the same way, after taking an m-th order derivative, we lose m derivatives of control on our output.

We first define the basic sobolev k-norm, for $k \in \mathbb{N}$, on $\mathbb{R}^n$. We will immediately generalize this to sections of a vector bundle eqiupped with a connection, but the underlying mechanism will be the same.

Definition 2.1. Define the *Sobolev k-norm* $\|\cdot\|_k$ on $C_c^\infty(\mathbb{R}^n)$ by

$$\|u\|_k^2 = \sum_{|\alpha|<k} \|\partial_x^\alpha u(x)\|_{L^2},$$

where α is a multi-index. Define $H^k(\mathbb{R}^n)$ to be the completion of $C_c^\infty(\mathbb{R}^n)$ under this norm.

Next, let E be a hermitian vector bundle with connection ∇ on a compact Riemannian manifold X. Given $u \in \Gamma(E)$, we have $\nabla u \in \sigma(T^*X \otimes E)$. Furthermore, using the tensor product connection on $T^*X \otimes E$, we have $\nabla\nabla u \in \sigma(T^*X \otimes T^*X \otimes E)$.

Definition 2.2. Define the *Sobolev k-norm* $\|\cdot\|_k$ on $\Gamma(E)$ by

$$\|u\|_k^2 = \sum_{j=0}^{k} \int_X |\nabla^k u|,$$

called the basic Sobolev k-norm on $\sigma(E)$.

The completion of $\sigma(E)$ under this norm is the Sobolev space $H^k(E)$.

The benefit of this definition is that it is inherently global, unlike some of the later definitions involving manifolds, where we have to verify independence of coordinates.

A fundamental fact about sobolev spaces is the following proposition:

Proposition 2.1. *A differential operator $P : \sigma(E) \to \sigma(E)$ of order m is bounded as a linear map $P : H^k \to H^{k-m}$. Hence, P extends to a linear map $P : H^k \to H^{k-m}$.*

To see why this is true, work in some coordinate patch. Write the connection locally as

$$\nabla_i = \frac{\partial}{\partial x_i} + \Gamma_i,$$

where Γ_i is a smooth local section of $\mathrm{End}(E)$. Then the k-norm of a section $u \in \Gamma(E)$ involves at most k ordinary derivatives of u, as a function on an open set of $\mathbb{R}^n$, plus many other lower order terms involving derivatives of u and the symbols Γ_i above. But fixing a covering of X by trivialized patches of E, we can bound these errors by some uniform constant as to not interfere with the end result.

2.0.1 The Fourier Transform

In fact, we can generalize to the spaces H^s, for $s \in \mathbb{R}$. To do this we need to take advantage of the Fourier Transform.

Notation: Here we will write $D_x^\alpha = -i\partial_x^\alpha$ for reasons which will become clear in the following proposition.

Definition 2.3. The *Fourier Transform* $\mathcal{F} : C_c^\infty(\mathbb{R}^n) \to C^\infty(\mathbb{R}^n)$ is defined to be the map $u \mapsto \hat{u}$, where

$$\hat{u}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-i\langle x,\xi\rangle} u(x)dx.$$

Here are some of its basic properties:

Proposition 2.2.

1. $\widehat{D_x^\alpha u(x)}(\xi) = \xi^\alpha \hat{u}(\xi)$
2. $\widehat{x^\alpha u(x)}(\xi) = D_\xi^\alpha \hat{u}(\xi)$

Proof:

1. Using integration by parts, and the fact that u is compactly supported
$$\widehat{D_x^\alpha u(x)}(\xi) = \int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle} D_x^\alpha u(x)dx = \int_{\mathbb{R}^n} D_x^\alpha e^{i\langle x,\xi\rangle} u(x)dx = \xi^\alpha \hat{u}(\xi).$$
2. Moreover,
$$\widehat{x^\alpha u(x)}(\xi) = \int_{\mathbb{R}^n} x^\alpha e^{i\langle x,\xi\rangle} u(x)dx = \int_{\mathbb{R}^n} D_\xi^\alpha e^{i\langle x,\xi\rangle} u(x) = D_\xi^\alpha \hat{u}(\xi).$$

In fact, $\mathcal{F}$ is an isometry $\mathscr{S} \to \mathscr{S}$, where $\mathscr{S}$ is the space of *Schwartz functions* of "rapidly decreasing functions," defined by

$$\mathscr{S} := \{u \in C^\infty(\mathbb{R}^n) : \forall \alpha, k, \exists C_{\alpha,k} \text{ such that } |D^\alpha u(x)| \le C_{\alpha,k}(1+|x|)^{-k}k, x \in \mathbb{R}^n\}$$

Its inverse is given by the following theorem:

Theorem 2.1. *For $u \in \mathscr{S}$, we have*

$$u(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle} \hat{u}(\xi)d\xi.$$

We will call the map on the right hand side the "inverse" Fourier transform.

2.0.2 Sobolev Spaces

Definition 2.4. We define the *Sobolev s-norm* $\|\cdot\|_s$ by the formula

$$\|u\|_s^2 = \int (1 + \|\xi\|)^{2s} |\hat{u}(\xi)|^2 d\xi.$$

To motivate this definition, note that for compactly supported u on $\mathbb{R}$, after integrating by parts, we have

$$\widehat{(\partial_x u)} = \int e^{ix\xi} \partial_x(u)(x)dx = \int \xi e^{ix\xi} u(x)dx = \xi \hat{u}.$$

Thus for the case $s = 1$, we have

$$\int |\xi|^2 |\hat{u}(\xi)|^2 d\xi = \int |\xi \hat{u}(\xi)|^2 d\xi = \int |\widehat{(\partial_x u)}(\xi)|^2 d\xi = \int |\partial_x u(x)|^2 dx$$

since $\mathcal{F}$ is an isometry in the L^2 norm.

One surprising result is the following:

Proposition 2.3 (The Sobolev Embedding Theorem). *Fix a positive integer k. Then for each real number $s > (n/2) + k$, there is a constant K_s such that for every $u \in \mathscr{S}$,*

$$\|u\|_{C^k} \leq K_s \|u\|_s.$$

Hence, there is a continuous embedding

$$H^s \subset C^k.$$

In order to extend these results globally to define Sobolev spaces on compact manifolds, we need a couple of lemmas, which we will not prove, but are not too unreasonable.

Lemma 2.1. *Let A be a smooth matrix-valued function on $\mathbb{R}^n$ so that $|D^\alpha A|$ is bounded for all α. Then the map $T : \mathscr{S} \to \mathscr{S}$ given by $Tu = Au$ extends to a bounded linear map $T : H^s \to H^s$ for all $s \in \mathbb{R}$.*

Lemma 2.2. *Let $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ be a C^∞ diffeomorphism which is linear outside a compact set. Then the map $T : \mathscr{S} \to \mathscr{S}$ given by $Tu = u \circ \Phi$ extends to a bounded linear map $T : H^s \to H^s$ for all $s \in \mathbb{R}$.*

Now we globalize. Let E be a smooth vector bundle over a compact manifold X. Fix a good presentation $\{U_\beta\}$ for E with coordinates $x_\beta : U_\beta \to \mathbb{R}^n$ and a partition of unity $\{\chi_\beta\}$ subordinate to the cover $\{B_\beta\} = \{x_\beta^{-1}(B^n)\}$.

Since $\sum \chi_\beta = 1$, any $u \in \Gamma(E)$ can be written $u = \sum u_\beta$ where $u_\beta = \chi_\beta u$. For $s \in \mathbb{R}$, define a Sobolev s-norm on $\Gamma(E)$ by setting

$$\|u\|_s = \sum_{\beta=1}^{N} \|u_\beta\|_s,$$

where the $\|u_\beta\|_s$ are defined in terms of the presentation $E|_{U_\beta} \simeq \mathbb{R}^n \times \mathbb{C}^n$.

As a consequence of 2.1 and 2.2, we have the following:

Proposition 2.4. *For any $s \in \mathbb{R}$ and any smooth vector bundle E over a compact manifold, the equivalence class of the norm $\|\cdot\|_s$ as defined above is independent of the good presentation used to define it. Furthermore, when $s = k$ is an integer, the equivalence class of $\|\cdot\|_k$ is the same as that defined in the first definition 2.2.*

Moreover, we have an analagous result to 2.1 for differential operators:

Proposition 2.5. *Let $P : \Gamma(E) \to \Gamma(F)$ be a differential operator of order m as defined in whose (locally expressed) coefficients are bounded as in 2.1. Then $P : \Gamma(E) \to \Gamma(E)$ extends to a bounded linear map $P : H^s(E) \to H^{s-m}(F)$ for all $s \in \mathbb{R}$.*

Remark. In fact, $H^s(E)$ is a hilbert space under the H^s inner product

$$\langle u, v\rangle_s = \int (1+|\xi|)^{2s}\hat{u}(\xi)\overline{\hat{v}(\xi)}d\xi.$$

Alternatively, we can define a pairing $\langle\cdot,\cdot\rangle_* : H^s(E)\times H^{-s}(E^*)\to\mathbb{R}$ by setting

$$\langle u, v\rangle_* = \int u(v),$$

placing $H^s(E)$ and $H^{-s}(E^*)$ in duality.

3 Pseudodifferential Operators

We start by giving a simple example to motivate the study of pseudodifferential operators. Let $P = \sum A^\alpha(x)D^\alpha$ be an ordinary differential operator. By the Fourier inversion formula, we have

$$u(x) = (2\pi)^{-n/2}\int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle}\hat{u}(\xi)d\xi.$$

Then applying P, we get

$$Pu(x) = P\left((2\pi)^{-n/2}\int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle}\hat{u}(\xi)d\xi\right) = \int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle}p(x,\xi)\hat{u}(\xi)d\xi, \tag{5}$$

where $p(x,\xi) = \sum A^\alpha(x)\xi^\alpha$. This $p(x,\xi)$ is called the *total symbol* of our differential operator

Example. Consider the operator $(1-\Delta) : \mathscr{S}\to\mathscr{S}$. Using the above technique, we can find its inverse A explicitly:

$$Au = (1-\Delta)^{-1}u = \int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle}\frac{1}{(1+\xi^2)}\hat{u}(\xi)d\xi.$$

We can check that

$$\begin{aligned}(1-\Delta)^{-1}(1-\Delta)u(x) &= \int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle}\frac{1}{(1+\xi^2)}\mathcal{F}((1-\Delta)u(x))(\xi)d\xi &(6)\\ &= \int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle}\frac{1}{(1+\xi^2)}(1-\xi^2)\hat{u}(\xi)d\xi &(7)\\ &= \int_{\mathbb{R}^n} e^{i\langle x,\xi\rangle}\hat{u}(\xi)d\xi = u(x). &(8)\end{aligned}$$

However, A is not a differential operator, since we cannot turn its symbol back into a polynomial in the $\left\{\frac{\partial}{\partial x_i}\right\}$'s. This suggests we broaden our focus to these new operators with more general symbols.

3.1 The Space of Symbols

For an ordinary differential operator of order m, its total symbol is an order m polynomial. Such an operator "removes" m orders of regularity, in the sense of our Sobolev spaces above. Naturally we would want to define of order m to behave in this manner as well. We will see that in fact the only factor which determines these new operators' continuity behavior in Sobolev spaces is the asymptotic polynomial behavior of the symbol.

Definition 3.1. Fix $m \in \mathbb{R}$. A smooth (matrix valued) function $p(x, \xi)$ on $\mathbb{R}^{2n}$ is called a *symbol of order m* if for each α, β, there is a constant $C_{\alpha,\beta}$ such that

$$|D_x^\alpha D_\xi^\beta p(x,\xi)| \leq C_{\alpha,\beta}(1+|xi|)^{m-|\alpha|}$$

for all x, ξ. Denote the space of these symbols by Sym_m.

Proposition 3.1. *To each $p \in Sym_m$ the formula 5 defines a linear operator $P : \mathscr{S} \to \mathscr{S}$. If p has compact x-support, this operator has a continuous extension $H^s \to H^{s-m}$ for all s.*

The space of all such operators is called the (or "a") space of pseudodifferential operators of order m on $\mathbb{R}^m$, denoted PDO_m.

We will formally define the *Distribution Kernel* of

References

[1] H. BLAINE LAWSON and MARIE-LOUISE MICHELSOHN. *Spin Geometry (PMS-38).* Princeton University Press, 1989.